

FIN630 Investment Analysis and Portfolio

Short Notes for Lecture 23-45

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INVESTMENT ANALYSIS & PORTFOLIO MANAGEMENT (Dr. Shahid A. Zia)

Lecture No.23

Random Walks Idea

- Does not state that security prices move randomly.
- Rather it maintains that the news arrives randomly.
- And in accordance with the EMH security prices rapidly adjust to this random arrival of news.

Consequences of Efficient Market

- Quick price adjustment in response to the arrival of random information makes the reward for analysis low.
- Prices reflect all available information.
- Price changes are independent of one another and move in a random fashion.
 - New information is independent of past.

Evidence on Market Efficiency

- Keys:
 - Consistency of returns in excess of risk.
 - Length of time over which returns are earned.
- Economically efficient markets:
 - Assets are priced so that investors cannot exploit any discrepancies and earn unusual returns.
 - Transaction costs matter.

Implications of Efficient Market Hypothesis

- What should investors do if markets are efficient?
- Technical analysis
 - Not valuable if the weak form holds.
- Fundamental analysis of intrinsic value.
 - Not valuable if semi-strong form holds.
 - Experience average results.
- For professional money managers
 - Less time spent on individual securities.
 - Passive investing favored.
 - Otherwise must believe in superior insight.
 - Tasks if markets informationally efficient
 - Maintain correct diversification.
 - Achieve and maintain desired portfolio risk.
 - Manage tax burden.
 - Control transaction costs.

Market Anomalies

- Exceptions that appear to be contrary to market efficiency.
- Earnings announcements affect stock prices.
 - Adjustment occurs before announcement but significant amount afterwards.
 - Contrary to efficient market because the lag should not exist.
 - The lag would then be a way of selling what you should bought earlier.
 - Extra returns than general public.
- Low P/E ratio stocks tend to outperform high P/E ratio stocks.
 - Low P/E stocks generally have higher risk-adjusted returns.
 - But P/E ratio is public information.
- Should portfolio be based on P/E ratios?
 - Could result in an undiversified portfolio.
- Size effect:
 - Tendency for small firms to have higher risk-adjusted returns than large firms.

- January effect:
 - Tendency for small firm stock returns to be higher in January.
 - Of 30% size premium, half of the effect occurs in January
- Value Line Ranking System:
 - Advisory service that ranks 1000 stocks from best (1) to worst (5)
 - Probable price performance in next 12 months.
 - 13 years study (1980-1993), Group 1 stocks had annualized return of 19%.
 - Best investment letter performance overall.
 - Transaction costs may offset returns.

Market Anomalies

- Profitability ratios
- Liquidity ratios
- Business Sales ratios

Conclusions About Market Efficiency

- Support for market efficiency is persuasive.
 - Much research using different methods.
 - Also many anomalies that cannot be explained satisfactorily.
- Markets very efficient but not totally.
 - To outperform the market, fundamental analysis beyond the norm must be done.
- If markets operationally efficient, some investors with the skill to detect a divergence between price and semi-strong value earn profits.
- Controversy about the degree of market efficiency still remains.
- Excludes the majority of investors.
- Anomalies offer opportunities.

Lecture No. 24

Financial Research

- Empirical Research
- Theoretical Research
- Behavioral Finance

Behavioral Finance

- Seeks to integrate psychology into the investment process.
- A variety of established behavior patterns may influence security prices.
- If market is going bullish we will find that people are gathering behind the same sentiments and thinking that the market is bullish whether it is or it isn't become secondary.
- Our behavior is also a function of how a problem is framed and the reference point we use in evaluating a situation.
- We may have definite preference for one alternative over another that, according to classical finance, is economically equivalent.
- People get carried away with sentiments.

Established Behaviors

- Loss Aversion
- Fear of Regret
- Myopic Loss Aversion
- Herding
- Anchoring
- Illusion of Control
- Prospect Theory
- Mental Accounting
- Asset Segregation
- Hindsight Bias
- Overconfidence
- Framing
- Illusion of Truth
- Biased Expectations
- Reference Dependence
- We also may misinterpret statistics, especially by misjudging the likelihood of events.

- Certain random numbers seem “less random” than others, and this belief influences certain investment decisions we might make.

Mistaken Statistics

- The Special Nature of Round Numbers
- Extrapolation
- Percentages vs. Numbers
- Apparent Order
- Regression to the Mean
- Sample Size

Risk Aversion

- People tend to react differently when confronted with losses rather gains.
- There is a tendency to become less risk averse or even risk seeking when an adverse event occurs.
- Investors may choose to gamble on an even bigger loss in the hope that the loss will disappear.

Equity Market Indicators

- Provide a composite report of market behavior on a given day.
- Dow Jones Industrial Average (DJIA).
 - Composed of 30 “blue-chip” stocks.
 - Price-weighted index: Essentially adds the prices of 30 stocks, divides by 30.
 - Adjusted for stock splits, stock dividends.
 - Oldest, most well-known measure.
- Standard & Poor’s Composite Index:
 - Composed of 500 “large” firm stocks.
 - Expressed as index number relative to a base index value of 10 (1941-43).
 - Value-weighted index: Prices and shares outstanding considered.
 - Indicates how much the average equity value of the 500 firms in the index has increased relative to the base period.
- NYSE and NASDAQ Composite Indices.
 - Value-weighted indices of broad markets.
- Nikkei 225 Average.
 - Price-weighted index of 225 actively-traded stocks on the Tokyo Stock Exchange.

Index

- Indexes are useful in assessing the performance of an investment.
- Choose which index:
 - It is important to ensure that the chosen index is an accurate proxy for what investors want to measure.
- A stock index should not be used with a bond portfolio.
- An index of large-capitalization stocks be used to judge a small-cap stock portfolio.
 - An investor can choose from any of the hundreds of indexes.
 - Equity securities:
- Dow Jones Industrial Average (DJIA)
- S&P 500
 - Financial research:
- S&P 500

Index Construction

- Price Weighting
- Equal Weighting
- Capitalization Weighting
- Volume based indices will tell you about companies that is best traded companies in the index.
- Price Weighting:
 - assigns heavy weight to high-priced stocks.
 - make use of a divisor to adjust for stock splits.
 - The stocks in a way in which you would understand that if you have Rs. 100 stock or Rs. 500 stock and Rs. 10 stock or Rs. 20 stock you would than give them, you would than perhaps add them up and divide by the number of stocks you want to based your indicator.
 - Capitalization Weighting:
 - considers the size of the company.
 - needs no adjustment for stock splits.
 - must be adjusted for changes in.

- index components.
- primary stock offerings.
- share repurchases.

Few Indexes

- ISE 10 index
- LSE 25 index
- KSE 100 index

LSE 25 INDEX

- LSE 25-Index includes the top 25 traded companies at LSE and captures 53% of the market capitalization and 98% of the total trading volume of LSE.

What is the KSE-100 Index?

- Highest market capitalization in each of the 34 sectors
- Largest market capitalization companies in descending order (66)

Lecture No. 25

Popular Indexes

- Stock Indexes
- Bonds Indexes
- Fixed Income Indexes
- International Indexes

What is the KSE-100 Index?

- A benchmark of the Equity Market.
- 100 Stocks listed on KSE.
- Represents 80 percent of the total market.
- Highest market capitalization in each of the 34 sectors.
- Largest market capitalization companies in descending order (66).

LSE 25 INDEX

- LSE launched a new 25-Index on December 20, 2002, which replaced the 101- Index. The Index has a Base Figure of 1000. (The Index closed at **5442.69** on 24th April, 2006).
- LSE 25-Index includes the top 25 traded companies at LSE and captures 53% of the market capitalization and 98% of the total trading volume of LSE.
- The Index was last reconstituted on July 1st 2006, in line with the regular review policy.
- KSE 100 is market capitalization based.
- LSE 25 is volume-based.

Methods of Investing

- Global Receipts
- Country Funds
- Individual Securities
- Unit Investment Trusts
- Local Mutual Fund
- International Mutual Fund
- Treasury bonds/bills

Marketable Financial Assets

- Commonly owned by individuals.
- Represent direct exchange of claims between issuer and investor.
- Usually very liquid or easy to convert to cash without loss of value.
- Examples: Savings accounts and bonds, certificates of deposit, money market deposit accounts.

Money Market Securities

- Marketable: claims are negotiable or salable in the marketplace.
- Short-term, liquid, relatively low risk debt instruments.
- Issued by governments and private firms.
- Examples: Money market mutual funds, T-Bills, Commercial paper.

Capital Market Securities

- Marketable debt with maturity greater than one year and ownership shares.
- More risky than money market securities.
- Fixed-income securities have a specified payment schedule.

- Dates and amount of interest and principal payments known in advance.

Securitization

- Transformation of illiquid, non-marketable risky individual loans into asset-backed securities.
 - GNMA's
 - Marketable securities backed by auto loans, credit-card receivables, small-business loans, leases.
- High yields, short maturities, investment-grade ratings.
- Risk free profits
- More riskier profit
- Investing for short term
- Investing for long term

Equity Securities

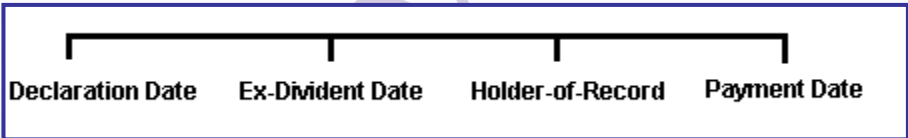
- Denote an ownership interest in a corporation.
- Denote control over management, at least in principle.
 - Voting rights important.
- Denote limited liability.
 - Investor cannot lose more than their investment should the corporation fail.

Preferred Stocks

- Behaves like bonds.
- Hybrid security because features of both debt and equity.
- Preferred stockholders paid after debt but before common stockholders.
 - Dividend known, fixed in advance.
 - May be cumulative if dividend omitted.
- Often convertible into common stock.
- May carry variable dividend rate.

Common Stocks

- Common stockholders are residual claimants on income and assets.
- Par value is face value of a share.
 - Usually economically insignificant.
- Market value is more significant.
- Book value is accounting value of a share in case a company goes bankrupt.
- Dividends are cash payments to shareholders.
 - Dividend yield is income component of return =D/P
 - Payout Ratio is ratio of dividends to earnings.



- Stock dividend is payment to owners in stock.
- Stock split is the issuance of additional shares in proportion to the shares outstanding.
 - The book and par values are changed.
- P/E ratio is the ratio of current market price of equity to the firm's earnings.

Investing Internationally

- Direct investing:
 - US stockbrokers can buy and sell securities on foreign stock exchanges.
 - Foreign firms may list their securities on a US exchange or on NASDAQ.
 - Purchase American Depositary Receipts (ADR's):
 - Issued by depositories having physical possession of foreign securities.
 - Investors isolated from currency fluctuations.
- It means if there is a stock of an emerging markets people or investors in developed markets might not be interested in investing in shares of an emerging market because of a fear of currency fluctuations.
- An asset back security / product based on foreign currencies shares but traded in your own currency.

Forms of Risks

- Country Risks / Sovereign Risks
- Political Risks
- Interest Rate Risks

- Economic Risks
- Regional Risks
- Risk in specific company
- Inflation Risks

In Upcoming Lecture

- Bonds
- Risk involved in trading
- Portfolio

Lecture No. 26

Bond Markets

- Secondary bond market is primarily an over-the-counter network of dealers.
 - NYSE features an automated bond system to execute orders similar to SuperDot & Instinet.
 - Mostly corporate bonds, thinly traded.
 - Treasury and agency bonds actively trade in dealers market.
 - Municipal bonds, local government bonds, city bonds & town bonds less actively traded.

Bond Characteristics

- Buyer of a newly issued coupon bond is lending money to the issuer who agrees to repay principal and interest.
- Bonds are fixed-income securities.
 - Buyer knows future cash flows.
 - Known interest and principal payments.
- If sold before maturity price will depend on interest rates at that time.
- Prices quoted as a % of par value.
- Bond buyer must pay the price of the bond plus accrued interest since last semiannual interest payment.
 - Prices quoted without accrued interest.
 - Premium payment
 - Discount payment
- Premium: amount above par value.
- Discount: amount below par value.

Innovation in Bond Features

- Zero-coupon bond:
 - Sold at a discount and redeemed for face value at maturity.
 - Locks in a fixed rate of return.
 - eliminating reinvestment rate risk.
 - Responds sharply to interest rate changes.

Major Bond Types

- Federal government securities (eg., T-bonds).
- Federal agency securities (e.g., GNMA's).
(Government National Mortgage Association)
- Federally sponsored credit agency securities (e.g., FNMA's).
(Federal National Mortgage Association)
- Municipal securities: General obligation bonds, Revenue bonds.

Corporate Bonds

- Usually unsecured debts maturing in 20-40 years, paying semi-annual interest, callable, with par value of \$1,000.
 - Callable bonds gives the issuer the right to repay the debt prior to maturity.
 - Convertible bonds may be exchanged for another asset at the owner's discretion.
 - Risk that issuer may default on payments

Bond Ratings

- Rate relative probability of default.
- Rating organizations:
 - Standard and Poors Corporation (S&P).
 - Moody's Investors Service Inc.
- Rating firms perform the credit analysis for the investor.
- Emphasis on the issuer's relative probability of default.
- Investment grade securities:
 - Rated AAA

- AA
- A
- BBB
- Typically, institutional investors are confined to bonds in these four categories
- Speculative securities:
 - Rated BB
 - B
 - CCC
 - CC
 - C
 - Significant uncertainties
 - C rated bonds are not paying interest

Securitization

- Transformation of illiquid, risky individual loans into asset-backed securities.
 - GNMA's.
 - Marketable securities backed by auto loans, credit-card receivables, small-business loans, leases.
- High yields, short maturities, investment-grade ratings.

Equity Securities

- Denote an ownership interest in a corporation.
- Denote control over management, at least in principle.
 - Voting rights important.
- Denote limited liability.
 - Investor cannot lose more than their investment should the corporation fail.

Preferred Stocks

- More like bonds.
- Hybrid security because features of both debt and equity.
- Preferred stockholders paid after debt but before common stockholders.
 - Dividend known, fixed in advance.
 - May be cumulative if dividend omitted.
- Often convertible into common stock.
- May carry variable dividend rate.

Common Stocks

- EPS
- Dividend
- Dividend Yield
- Dividend Payout Ratio
- P/E Ratio

Lecture No. 27

Alternative Investments Non-marketable Financial Assets

- Usually very liquid or easy to convert to cash without loss of value.
- Examples: Savings accounts and bonds, certificates of deposit, money market deposit accounts.

Money Market Securities

- Marketable: claims are negotiable or salable in the marketplace.
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Derivative Securities

- Securities whose value is derived from another security or derived from an underlying assets.
- Futures and options contracts are standardized and performance is guaranteed by a third party.
- Warrants are options issued by firms.
- Risk management tools.

Market Mechanics

- Regular Markets
- Spot Markets
- Original Markets
- Future Markets

Options

- Exchange-traded options are created by investors, not corporations.
- Call (Put): Buyer has the right but not the obligation to purchase (sell) a fixed quantity from (to) the seller at a fixed price before a certain date.
 - Right is sold in the market at a price.
- Increases return possibilities.
- Chicago Board of Auction Exchange.

Derivatives

- Warrants
- Options
- SWOPS

Futures

- Futures contract: A standardized agreement between a buyer and seller to make future delivery of a fixed asset at a fixed price.
 - A “good faith deposit,” called margin, is required of both the buyer and seller to reduce default risk.
 - Used to hedge the risk of price changes.

Bond Fundamentals Bond Principles

- Identification of Bonds
- Classification of Bonds
- Terms of Repayment
- Bond Cash Flows
- Convertible and Exchangeable Bonds
- Registration
- Bonds are identified by:
 - Issuers
- Coupon
- Maturity Year

Who is behind the Bond / Issuer

Coupon Rate

Maturity Period

Classification of Bonds

- Bonds are classified according to:
 - the nature of the issuer.
 - Security behind the bonds.
 - Some bonds provide a conversion feature.
 - exchanged for another asset.
 - usually shares of common stock in the issuing companies.
 - The bond indenture spells out the details.

Terms of Repayment

- The income stream associated with most bonds contains:
 - an annuity stream
 - a single sum to be received in the future
 - Semi-annual
- Bond Pricing & Returns
- Valuation Equations

- Yield to Maturity
- Spot Rates
- Realized Compound Yield
- Current Yield
- Accrued Interest

Dividend Reinvestment Program

Yield to maturity

- The discount rate that equates the present value of the future cash flows with the current price of the bond.
- By tradition, bond yield to maturity is based on semiannual compounding.

Bond Cash Flows

- A major assumption of the yield to maturity calculation:
 - the requirement that coupon proceeds be reinvested at the bond's yield to maturity.
- If the reinvestment rate is different from the bond's rate, the rate of return ultimately realized will be different.

Example

Return of Bond

- When comparing bonds with other investments, the effective annual yield (realized compound yield) should be used to make a realistic comparison.
- The yield curve shows the relationship between yield and time until maturity.
- Bonds accrue interest each day they are held.

Bond Prices

- Bond prices are expressed as a percentage of par value.
- Corporate bonds usually trade in minimum price increments of 1/8%.
- Government bonds trade in 1/32nds.

Lecture No. 28

Investing Internationally

- **Direct investing:**
 - US stockbrokers can buy and sell securities on foreign stock exchanges.
 - Foreign firms may list their securities on a US exchange or on NASDAQ.
 - Purchase ADR's
 - Purchase GDR's

Derivative Securities

- Securities whose value is derived from another security.
- Futures and options contracts are standardized and performance is guaranteed by a third party.
 - Risk management tools.
- Warrants are options issued by firms.

Bond Pricing & Returns

- Valuation Equations
- Yield to Maturity
- Spot Rates
- Realized Compound Yield
- Current Yield
- Accrued Interest

Bond Risks

Price Risks

- Price Risks Refer to the chance of monetary loss due to:

1. default risk:

The likelihood of the firm defaulting on its loan repayments.

2. interest rate risk:

The variability of interest rates.

Sovereign Risk

Convenience Risks

- Refer to additional demands on management time because of;
 - Bonds being called by their issuers.
 - The need to reinvest interest received.
 - Poor marketability of a particular issue.

- May bonds have a period of call protection and subsequently a declining call premium.

Bonds

Interest Rates

- Rates and basis points:
 - 100 basis points are equal to one percentage point.
- Short-term riskless rate:
 - Provides foundation for other rates.
 - Approximated by rate on Treasury bills.
 - Other rates differ because of;
 - Maturity differentials
 - Security risk premiums
- Maturity differentials:
 - Term structure of interest rates.
 - Accounts for the relationship between time and yield for bonds the same in every other respect.
- **Risk premium:**
 - Yield spread or yield differential.
 - Associated with issuer's particular situation.

Determinants of Interest Rates

- Real rate of interest:
 - Rate that must be offered to persuade investors to save rather than consume.
 - Rate at which real capital physically reproduces itself.
- Nominal interest rate:
 - Function of the real rate of interest and expected inflation premium.

Determinants of Interest Rates

- Market interest rates on interest debt $\approx$
- real rate + expected inflation
 - Fisher Hypothesis
 - Real Estate
- Real rate estimates obtained by subtracting the expected inflation rate from the observed nominal rate.

Bond Pricing Relationships

- Inverse relationship between price and yield.
- An increase in a bond's yield to maturity results in a smaller price decline than the gain associated with a decrease in yield.
- Long-term bonds tend to be more price sensitive than short-term bonds.
- The relationships with respect to maturity are not exact as they are when duration is used.
- In discussing the pricing relationships it is helpful to discuss how maturity and cash flows as measured by coupon rates must be considered to get exact relationships.
- As maturity increases, price sensitivity increases at a decreasing rate.
- Price sensitivity is inversely related to a bond's coupon rate.
- Price sensitivity is inversely related to the yield to maturity at which the bond is selling.

Measuring Bond Yields

- Yield to maturity:
 - Most commonly used.
 - Promised compound rate of return received from a bond purchased at the current market price and held to maturity.
 - Equates the present value of the expected future cash flows to the initial investment.
 - Similar to internal rate of return.

Yield to Maturity

$$P = \sum_{t=1}^{2n} \frac{C_t/2}{(1 + YTM/2)^t} + \frac{MV}{(1 + YTM/2)^{2n}}$$

$$YTM = 2 \times \{[MV/P]^{1/2n} - 1\}$$

- Solve for YTM:
 - For a zero coupon bond
- Investors earn the YTM if the bond is held to maturity and all coupons are reinvested at YTM

Yield to Call

- Yield based on the deferred call period.
- Substitute number of periods until first call date for and call price for face value.

$$P = \sum_{t=1}^{2c} \frac{C_t/2}{(1+YTC/2)^t} + \frac{CP}{(1+YTC/2)^{2c}}$$

Realized Compound Yield

- Rate of return actually earned on a bond given the reinvestment of the coupons at varying rates

$$RCY = \left[\frac{\text{Total future dollars}}{\text{Purchase price of bond}} \right]^{1/2n} - 10$$

- Horizon return analysis
 - Bond returns based on assumptions about reinvestment rates
 - Rates will vary.

Bond Valuation Principle

- Intrinsic value:
 - An estimated value.
 - Present value of the expected cash flows.
 - Required to compute intrinsic value;
 - Expected cash flows.
 - Timing of expected cash flows.
 - Discount rate, or required rate of return by investors.

Bond Valuation

- Value of a coupon bond:

$$V = \sum_{t=1}^{2n} \frac{C_t/2}{(1+r/2)^t} + \frac{MV}{(1+r/2)^{2n}}$$

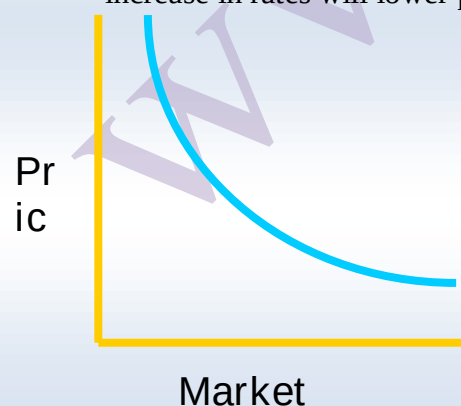
- Biggest problem is determining the discount rate or required yield.
- Required yield is the current market rate earned on comparable bonds with same maturity and credit risk.

Bond Price Changes

- Over time, bond prices that differ from face value must change.
- Bond prices move inversely to market yields.
- The change in bond prices due to a yield change is directly related to time to maturity and indirectly related to coupon rate.

Bond Price Changes

- Holding maturity constant, a rate decrease will raise prices a greater percentage than a corresponding increase in rates will lower prices.



Lecture # 29

Bond Risks

Price Risks

- Interest Rate Risk

- Default Risk
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1. Default Risk:

The likelihood of the firm defaulting on its loan payments.

2. Interest Rate Risk:

The variability of interest rates.

Convenience Risks

- Refer to additional demands on management time because of;
 - bonds being called by their issuers.
 - the need to reinvest interest received.
- May bonds have a period of call protection and subsequently a declining call premium.
- Poor marketability of a particular issue.

Interest Rates

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 - 100 basis points are equal to one percentage point.
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Determinants of Interest Rates

- Real rate of interest:
 - Rate that must be offered to persuade individuals to save rather than consume.
 - Rate at which real capital physically reproduces itself.
- Nominal interest rate:
 - Function of the real rate of interest and expected inflation premium.
- Market interest rates on risk less debt $\approx$ real rate + expected inflation
 - Fisher Hypothesis

Bond Pricing Relationships

- Inverse relationship between price and yield.
- An increase in a bond's yield to maturity results in a smaller price decline than the gain associated with a decrease in yield.
- As maturity increases, price sensitivity increases at a decreasing rate.
- Price sensitivity is inversely related to a bond's coupon rate.
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$$P = \sum_{t=1}^{2n} \frac{C_t/2}{(1+YTM/2)^t} + \frac{MV}{(1+YTM/2)^{2n}}$$

- For a zero coupon bond:

$$YTM = 2 \times \{[MV/P]^{1/2n} - 1\}$$

- Investors earn the YTM if the bond is held to maturity and all coupons are reinvested at YTM.

Yield to Call

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Realized Compound Yield

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- Horizon return analysis:
 - Bond returns based on assumptions about reinvestment rates.

Bond Valuation Principle

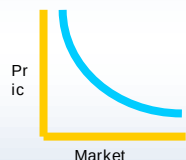
- **Intrinsic value:**
 - An estimated value.
 - Present value of the expected cash flows.
 - Expected cash flows
 - Ratings
 - Timing of expected cash flows

Bond Valuation

- Value of a coupon bond:
Formula:
- Biggest problem is determining the discount rate or required yield.
- Required yield is the current market rate earned on comparable bonds with in the same maturity and credit risk.

Bond Price Changes

- Over time, bond prices that differ from face value must change.
- Bond prices move inversely to market yields.
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- Holding maturity constant, a rate decrease will raise prices a greater percent than a corresponding increase in rates will lower prices.



Measuring Bond Price Volatility: Duration

- Important considerations:
 - Different effects of yield changes on the prices and rates of return for different bonds.
 - Maturity
 - It May not have an identical economic lifetime.
 - A measure is needed that accounts for both size and timing of cash flows.
 - Maturity is an inadequate measure of volatility.

What is Duration

- It is a measure of a bond's lifetime, stated in years, that accounts for the entire pattern (both size and timing) of the cash flows over the life of the bond.
- The weighted average maturity of a bond's cash flows.
 - Weights determined by present value of cash flows.
- A measure of the effective maturity of a bond.
- The weighted average of the time until each payment is received, with the weights proportional to the present value of the payment.
- Duration is shorter than maturity for all bonds except zero coupon bonds.
- Duration is equal to maturity for zero coupon bonds.

- The description of duration that is used here stresses the concept of average life.
- Since the measurement of duration considers the timing and value of intermediate payments.
- It is an accurate measure of average life and is more meaningful than maturity for any bond than has coupon payments.

Why is Duration Important?

- It allows comparison of effective lives of bonds that differ in maturity, coupon.
- It is used in bond management strategies particularly immunization.
- Measures bond price sensitivity to interest rate movements, which is very important in any bond analysis.

Duration/Price Relationship

- Price change is proportional to duration and not to maturity.
- One of the key properties of duration is related to price changes.
- Price changes on fixed-income securities are proportional to duration.
- Duration incorporates both the coupon rate and maturity effects into a single measure.
- The concept of modified duration is used extensively in industry.

Duration

- Duration is an average signifying the point in time when the PV of a security is repaid.
- A measure of the price sensitivity of a security to interest rate changes.
- $(\Delta P)/P = -D \times (\Delta i)/(1+i) \approx -D \Delta i$
- Measure of bond volatility.
- Holding period sufficient to balance price and reinvestment risk assuring the investor the yield to maturity.
- Ratio of sum of discounted, time weighted CF divided by price of security.
- Duration values change each day.
- Measure of interest rate risk.
 - as measure of bond volatility.
- Inversely related to coupon rate & current market rate of return.
- More refined measure of maturity that helps determine how much more risk there is.
- Weighted average of time it takes to recoup your investment - counts interest (coupon) payments as well as principal payment.
- Each payment is discounted in terms of its PV.
- For zero coupon bonds, duration = maturity - get all CF at maturity.
- Key concepts.
- The longer the maturity of a bond, (other things equal), the greater its price volatility.
- The smaller the coupon of the bonds (other things equal) the greater its Duration & the greater its price sensitivity (volatility).
- Duration is a positive function of term to maturity & a negative function of size of the coupon of the bond (extremely low coupon - zero coupon, duration = maturity).
- Numerator is PV of all cash flows weighted according to length of time to receipt.
- Denominator = PV of cash flows = Price
- For zero coupon security, duration = maturity
- With longer duration, need a larger price change to get same yield change - can see that in Treasury Bond, Note & Bill column, 30-year bond fell (10 / 32) on 7/16/98, while 2 yr note fell only (1 / 32).

Uses of Duration

- Maturity Matching approach
- Zero coupon approach
- Duration-Matching approach
- Duration is a measure of interest rate risk that considers the effects of both the coupon rate & maturity changes in bond prices.
- By matching the duration of their firm's assets & liabilities, managers of financial institutions can minimize exposure to interest rate risk.
- Duration is a measure of interest rate risk that considers the effects of both the coupon rate & maturity changes in bond prices.
- By matching the duration of their firm's assets & liabilities, managers of financial institutions can minimize exposure to interest rate risk.

- Duration is the figure that we need to look at when we talking about being able to measure interest rate risk and by looking at the duration will be able to avoid the effect of interest rate risk on our realized rates of return.
- Immunization of interest rate risk is a tool that can be used for passive management.
- Financial institutions use immunization concept to manage assets and liabilities.
- To control for interest rate risk, managers of financial institutions balance the durations of their asset and liability portfolios.
- Target date immunization can be used to lock in a fixed rate of return for some investment horizon.

Duration problem

Example

- **Example:** \$100 par, 6 % coupon, 5 years (10 semi-annual payments of 3% each six months) in an 8 % market (4 % semi-annual yield).

Per (n)	CF	DF	PV of CF	n x PV of CF
1	\$ 3.0	.9615	2.8845	2.8845
2	3.0	.9246	2.7738	5.5476
3	3.0	.8890	2.667	8.001
4	3.0	.8548	2.5664	10.2576
5	3.0	.8219	2.4657	12.3285
.....				
10	103.0	.6756	69.8568	698.568
Per (n)	CF	DF	PV of CF	nx PV of CF
1	3	0.9615	2.8845	2.8845
2	3	0.9246	2.7738	5.5476
3	3	0.889	2.667	8.001
4	3	0.8548	2.5644	10.2576
5	3	0.8219	2.4657	12.3285
6	3	0.7903	2.3709	14.2254
7	3	0.7599	2.2797	15.9579
8	3	0.7307	2.1921	17.5368
9	3	0.7026	2.1078	18.9702
10	103	0.6756	69.8568	698.568
Sum PV & weighted PV column 91.8927 801.5775				

- Duration of this security = $801.5775 / 91.8927 = 8.7229726$ half years
 - = 4.36 years
- So the five year 6% coupon security has a duration of 4.36 years in a market where expected yield to maturity = 8%
- Find duration if i = 4 %
- Is it the same as when i = 8 % ? (hint: no!)
- Modified duration = duration / (1 + i)

Lecture # 30

Rules for Duration

- Rule 1: The duration of a zero-coupon bond equals its time to maturity.
- Rule 2: Holding maturity constant, a bond's duration is higher when the coupon rate is lower.
- Rule 3: Holding the coupon rate constant, a bond's duration generally increases with its time to maturity.
- Rule 4: Holding other factors constant, the duration of a coupon bond is higher when the bond's yield to maturity is lower.

Duration Conclusions

- To obtain maximum price volatility, investors should choose bonds with the longest duration.
- Duration is additive.
 - Portfolio duration is just a weighted average.

- Duration measures volatility which isn't the only aspect of risk in bonds.
- Default risk in bonds.
- Interest rate risks in bonds.

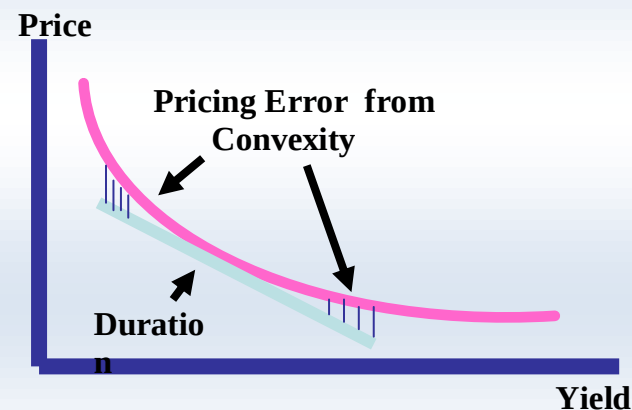
Convexity

- Convexity is very important consideration in the bond market.
- Significance of convexity.

Convexity

- Refers to the degree to which duration changes as the yield to maturity changes.
 - Price-yield relationship is convex
- Duration equation assumes a linear relationship between price and yield.
- Convexity largest for low coupon, long-maturity bonds, and low yield to maturity.

Pricing Error from Convexity



Bond price volatility

- Bond prices are inversely related to bond yields.
- The price volatility of a long-term bond is greater than that of a short-term bond, holding the coupon rate constant.
- The price volatility of a low-coupon bond is greater than that of a high-coupon bond, holding maturity constant.

Example of Bond volatility

- The price of a 2-year bond (with one year remaining - so really a 1-year bond) fell by 9.57% when market yield increased from 5 % to 6% and rose by 9.71% when market yields fell from 5% to 4%.
- This measure of percentage price change is called bond volatility.
- What if we had a bond with 30 years remaining & a 5% coupon and market rates rose to 6%.
- Again assume semi-annual coupons of 2.5% when market semi-annual yields are 3%.

Bond volatility

- $P = 25/(1.03) + 25/(1.03)^2 + 25/(1.03)^3 + 25/(1.03)^4 + 25/(1.03)^5 + \dots 25/(1.03)^{30} + 1000/(1.03)^{30}$
 $= 24.27 + 23.56 + 22.88 + \dots + 4.24 + 169.73 = 861.62$
- What is the percentage change in price?
- $\% \Delta P = (861.62 - 1000)/1000 = - 98/1000 = - 13.8\%$
- For the same change in yield (from 5% to 6%) the 1-year security had a 9.57% change in price & the 30-year security had a 13.8% change in price.

Bond price volatility

- Bond price volatility is the percentage change in bond price for a given change in yield.
- Volatility increases with maturity.
- A 1% increase in market yields from 5% gives volatility of;
- 1 year - 9.57% (price fell to 990.43)
- 15 year - 9.8% (price fell to 902)
- 30 year - 13.8% (price fell to 861.62)

Zero Coupon

- What is the volatility of a 30-year zero coupon bond if market rates change from 5% to 6%
- $P_0 = 1000 / (1.05)^{30} = 231.38$
- $P_1 = 1000 / (1.06)^{30} = 174.11$
- $\% \Delta P = (174.11 - 231.38)/231.38 = - 32.9\%$
- Conclusion: lower coupon has more volatility

(- 32.9 % vs. -13.8% for 5% coupon)

- If assume 60-semi annual compounding periods with $c = I = 2.5\%$, $P_0 = 1000 / (1.025)^{60} = 227.28$
- or $P_1 = 1000 / (1.03)^{60} = 169.73$
- $\% \Delta P = (169.73 - 227.28) / 227.28 = -33.9\%$

Summary

- Bond prices are inversely related to bond yields.
- The price volatility of a longer-maturity security is larger than that of a shorter maturity security, holding the coupon rate constant.
- The price volatility of a low coupon security is larger than that of a high coupon security, holding maturity constant.
- IRR (Interest rate risk) occurs when realized return doesn't = YTM
 - Why does this happen? How can it be avoided?

What does the price volatility of a bond depend on?

1. Maturity
2. Coupon
3. Yield to maturity

Approximate % change in price of bond

- To get a approximate % change in the price of a bond for a change in yield we just multiply it by the change in yield.
- This is a linear approximation; good for small change in interest rates.

Lecture # 31

Bonds

- So now you've learned the basics of bonds. Not too complicated, was it?
- Here is a recap of what we discussed:
- Bonds are just like IOUs.
- Buying a bond means you are lending out your money.
- Bonds are also called fixed-income securities because the cash flow from them is fixed.
- Stocks are equity; bonds are debt.
- Issuers of bonds are governments and corporations.
- A bond is characterized by its face value, coupon rate, maturity, and issuer.
- Yield is the rate of return you get on a bond.
- When price goes up, yield goes down and vice versa.
- Interest Rates
- When interest rates rise, the price of bonds in the market falls.
- When interest rates go down, the price of bonds in the market rise.
- Bills, notes, and bonds are all fixed-income securities classified by maturity.
- Government bonds are the safest, followed by municipal bonds, and then corporate bonds.
- Bonds are not risk free. It's always possible especially for corporate bonds, for the borrower to default on the debt payments.
- High risk / high yield bonds are known as junk bonds.

Understanding

Risk and Return

- Two key concepts in Finance are:
 - The time value for money
 - The fact that a safe rupee is worth more than a risky rupee.
- The trade-off is the central theme in the investment decision-making process.

Return

- Holding Period Return
- Yield and Appreciation
- Compounding
- Compound Annual Return
- Simple interest / returns

Holding Period Return

- Independent of the passage of time.
- Should only be used to compare investments over identical time periods.
- Consider the yield of an investment from interest or dividends.
- Consider the appreciation from a change in the investment value.

Supplement Their Income

- Returns in the form of Interest
- Returns in the form of Dividend

Time Value of Money

- Its calculations overcome the shortcomings of the holding period return.
- Permit a direct comparison between a particular sum today and amounts in the future.
- Inflation
- Worth investing or not
- Compound annual return - the interest rate that satisfies a time value of money equation.
- The number of compound periods per year can significantly increase the compound annual return.
- Semi-annual coupon rates
- Twice in year payments
- Multiples of coupon rates
- Purchase of bond in January
- Purchase of bond in April

Hypothetical Assumption

- 2 biannual or semi-annual return
- 4 quarterly returns to be reinvested

Risk

- A chance of loss.
- Inseparable from time.
- Breakeven – the point decide gain or loss.
- Virtually all investors are risk averse, especially with significant sums of money. Total risk – complete variability of investment results.
- Total risk can be partitioned into diversifiable and un-diversifiable risk.
- The marketplace only rewards un-diversifiable risk.
- Risk is unavoidable.
- A direct relationship between expected return and unavoidable risk.
- Risk investment doesn't guarantee a return.
- Unnecessary risk doesn't warrant any additional return.

Breakeven

- Cost
 - Manufacturing cost
 - Financial cost
 - Administrative cost
 - Marketing cost

Factors

- Greed
- Fear

Junk Bonds

Risk

- Total risk is complete variability of investment results.
- Total risk can be partitioned into diversifiable and un-diversifiable risk.
- The marketplace only rewards un-diversifiable risk.
- No Pain No Gain
- Risk is unavoidable
- A direct relationship between expected return and unavoidable risk.
- Risk investment doesn't guarantee a return.
- Unnecessary risk doesn't warrant any additional return.

Lecture # 32

So, What is Risk?

- Whether it is [investing](#), driving, or just walking down the street, everyone exposes themselves to risk.
- Risk is the chance that an [investment's](#) actual return will be different than expected.
- This includes the possibility of losing some or all of the original investment.
- When [investing in stocks](#), bonds, or any investment instrument there is a lot more risk than you'd think.
Let's examine closer the [different types of risk](#).

Risk Types

- Two general types;
 - Systematic Risk
 - Non-systematic Risk
- Systematic (general) Risk is a risk that influences a large number of assets.
 - An example is political events. It is virtually impossible to protect yourself against this type of risk.
 - Pervasive, affecting all securities, cannot be avoided
 - Interest rate or market or inflation risks
- Non-systematic (specific) Risk is Sometimes referred to as "specific risk". It's risk that affects a very small number of assets.
- An example is news that affects a specific stock such as a sudden strike by employees.
- **Total Risk** = Systematic (General) Risk + Non-systematic (Specific) Risk
- **Diversification** is the only way to protect yourself from unsystematic risk.

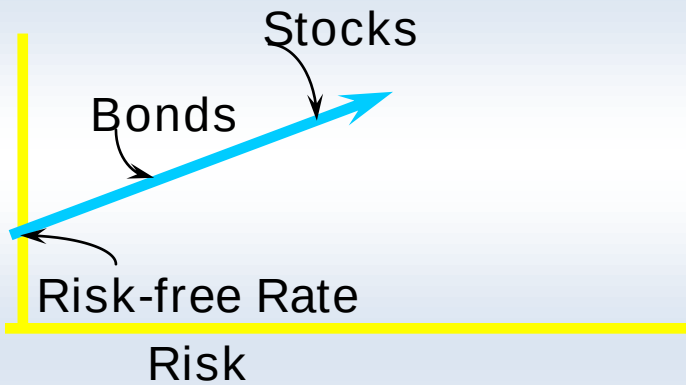
Risk Sources

- Interest Rate Risks
- Market Risk
- Inflation Risk
- Business Risk
- Financial Risk
- Liquidity Risk
- Exchange Rate Risk
- Country Risk
- Liquidity Risk
 - Marketability with-out sale prices
- Financial Risk
 - Tied to debt financing
- Inflation Risk
 - Purchasing power variability
- **Credit or Default Risk:**
- This is the risk that a company or individual will be unable to pay the contractual interest or principal on its debt obligations.
- This type of risk is of particular concern to investors who hold bond's within their portfolio.
- Government [bonds](#), especially those issued by the Federal government, have the least amount of default risk and least amount of returns while corporate bonds tend to have the highest amount of default risk but also the higher [interest rates](#).
- **Country Risk:**
- This refers to the risk that a country won't be able to honor its financial commitments. When a country defaults it can harm the performance of all other financial instruments in that country as well as other countries it has relations with.
- **Interest Rate Risk:**
- A rise in interest rates during the term of your debt securities hurts the performance of stocks and bonds.
- **Political Risk:**
- This represents the financial risk that a country's government will suddenly change its policies.
- This is a major reason that second and third world countries lack foreign investment.
- **Market Risk** - This is the most familiar of all risks.
- It's the day to day fluctuations in a stocks price.
- Also referred to as [volatility](#).
- Market risk applies mainly to stocks and options. As a whole, stocks tend to perform well during a bull market and poorly during a bear market—volatility is not so much a cause but an effect of certain market forces.

The Tradeoff Between

ER and Risk

- Investors manage risk at a cost – means lower expected returns (ER).
- Any level of expected return and risk can be attained.



The Return and Risk from Investing: Asset Valuation

- AV is a Function of both return and risk
 - It is at the center of security analysis
- How should realized return and risk be measured?
- The realized risk-return
- The expected risk-return
 - The **realized** risk-return tradeoff is based on the past
 - The **expected** risk-return tradeoff is uncertain and may not occur

Return Components

- Returns consist of two elements:
 - Periodic Cash Flows
 - Price Appreciation or Depreciation
 - **Periodic cash flows** such as interest or dividends (income return).
 - “Yield” measures relate income return to a price for the security.
 - **Price appreciation or depreciation** (capital gain or loss).
 - The change in price of the asset.
- **Total Return** = Yield + Price Change

Arithmetic Versus Geometric

- Arithmetic mean does not measure the compound growth rate over time.
 - It does not capture the realized change in wealth over multiple periods.
 - But it does capture typical return in a single period.
- Geometric mean reflects compound, cumulative returns over more than one period

Risk Premiums

- Premium is additional return earned or expected for additional risk
 - Calculated for any two asset classes
- Equity Risk Premium
- Bond Horizon Premium
- **Equity risk premium** is the difference between stock and risk-free returns.
- **Bond horizon premium** is the difference between long- and short-term government securities.
- Equity Risk Premium, (ERP) =

$$\left[\frac{(1 + TR_{CS})}{(1 + RF)} \right] - 1$$

- Bond Horizon Premium, (BHP) =

The Risk-Return Record

- Since 1920, cumulative wealth indexes show stock returns dominate bond returns.
 - Stock standard deviations also exceed bond standard deviations.
- Annual geometric mean return for the S&P 500 is 10.0% with standard deviation of 19.4%

Measuring Returns

- We measure for comparing performance over time or across different securities.
- Total Return is a percentage relating all cash flows received during a given time period, (denoted CF_t + (PE - PB), to the start of period price, PB)

$$TR = \frac{CF_t + (P_E - P_B)}{P_B}$$

Measuring Returns

- We measure for comparing performance over time or across different securities.
- Total Return (TR) can be either positive or negative.
 - When cumulating or compounding, negative returns are problem.
- A Return Relative (RR) solves the problem because it is always positive.

$$RR = \frac{CF_t + P_E}{P_B} = 1 + TR$$

- To measure the level of wealth created by an investment rather than the change in wealth, we need to cumulate returns over time.
- Cumulative Wealth Index, CWIn, over n periods =

$$WI_0(1+TR_1)(1+TR_2)...(1+TR_n)$$

- To measure the level of wealth created by an investment rather than the change in wealth, we need to cumulate returns over time.
- International returns include any realized exchange rate changes.
 - If foreign currency depreciates, returns are lower in domestic currency terms.
 - If foreign currency appreciates, returns in local currency shall also grow.

Measuring International Returns

- International returns include any realized exchange rate changes.
 - If foreign currency depreciates, returns are lower in domestic currency terms.
- Total Return in domestic currency =

$$\left[RR \times \frac{\text{End Val. of For. Curr.}}{\text{Begin Val. of For. Curr.}} \right] - 1$$

Measures Describing a Return Series

- TR (Total Return), RR (Return Relative) and CWI (Cumulative Wealth Index) are useful for a given, single time period.
- What about summarizing returns over several time periods?
- Arithmetic mean, or simply mean,

$$\bar{X} = \frac{\sum X}{n}$$

Geometric Mean

- Defined as the n-th root of the product of n return relatives minus one or G =
- Difference between Geometric mean and Arithmetic mean depends on the variability of returns,

Adjusting Returns for Inflation

- Returns measures are not adjusted for inflation.
 - Purchasing power of investment may change over time.
 - Consumer Price Index (CPI) is possible measure of inflation.

Measuring Risk

- Risk is the chance that the actual outcome is different than the expected outcome.
- Standard Deviation measures the deviation of returns from the mean.

Lecture # 33

Investment Decisions

- Underlying investment decisions: the tradeoff between expected return and risk.
 - Expected return is not usually the same as realized return.
- Risk: the possibility that the realized return will be different than the expected return.
- The risk / return tradeoff could easily be called the iron stomach test. Deciding what amount of risk you can take on while allowing you to get rest at night is an investor's most important decision.

The Investment Decision Process

- Two-step process:

- Security analysis and valuation.
 - Necessary to understand security characteristics.
- Portfolio management:
 - Selected securities viewed as a single unit.
 - How efficient are financial markets in processing new information?
 - How and when should it be revised?
 - How should portfolio performance be measured?

Basic Strategies

- The basic decision involved in fixed-income management is the decision to be active or passive.
- An active strategy has as its goal to **secure superior returns from the fixed-income portfolio.**
- Superior returns can be earned if the investor can predict interest rate movements that are not currently incorporated into price or if the investor can identify bonds that are mis-priced for other factors.
- For example, finding a bond that has a credit risk premium that is too large for its credit risk.
- Passive management involves controlling risk and balancing risk and return.

Fixed Income

Passive Management

- Bond-Index Funds
- Immunization of interest rate risk.
 - Net worth immunization:

Duration of assets = Duration of liabilities

- Target date immunization:

Holding Period matches Duration

- Cash flow matching and dedication.
- Bonds growth

Bond Index Fund

Contingent Immunization

- Combination of active and passive management.
- Strategy involves active management with a floor rate of return.
- As long as the rate earned exceeds the floor, the portfolio is actively managed.
- Once the floor rate or trigger rate is reached, the portfolio is immunized.

Factors Affecting the Process

- Uncertainty in *ex post* returns dominates decision process.
 - Future unknown and must be estimated.
- Foreign financial assets: opportunity to enhance return or reduce risk.
- Quick adjustments are needed to a changing environment.
- The Internet and the investment opportunities.
- Institutional investors are important.

Portfolio theory: Investment Decisions

- Involve uncertainty
- Focus on expected returns
 - Estimates of future returns needed to consider and manage risk.
- Goal is to reduce risk without affecting returns.
 - Accomplished by building a portfolio.
 - Diversification is key.

Dealing With Uncertainty

- Risk that an expected return will not be realized.
- Investors must think about return distributions, not just a single return.
- Probabilities weight outcomes:
 - Can be discrete or continuous.
 - Should be assigned to each possible outcome to create a distribution.
 - It helps to diversify.

Calculating Expected Return

- Expected value:
 - The single most likely outcome from a particular probability distribution.
 - The weighted average of all possible return outcomes.
 - Referred to as an *ex ante* or expected return.

Calculation A Risk

- Variance and standard deviation used to quantify and measure risk.
 - Measures the spread in the probability distribution.
 - Variance of returns: $\sigma^2 = \sum (R_i - E(R))^2 p_i$
 - Standard deviation of returns:
 $\sigma = (\sigma^2)^{1/2}$
 - Ex ante rather than ex post σ relevant.

Portfolio Expected Return

- Weighted average of the individual security expected returns.
 - Each portfolio asset has a weight, w , which represents the percent of the total portfolio value.

Portfolio Expected Return

Asset Management

Portfolio Risk

- Portfolio risk not simply the sum of individual security risks.
- Emphasis on the rate of risk of the entire portfolio and not on risk of individual securities in the portfolio.
- Individual stocks are risky only if they add risk to the total portfolio

Example

- Measured by the variance or standard deviation of the portfolio's return.
 - Portfolio risk is not a weighted average of the risk of the individual securities in the portfolio.

$$\sigma_p^2 \neq \sum_{i=1}^n w_i \sigma_i^2$$

Lecture # 34

Risk Reduction in Portfolios

- Assume all risk sources for a portfolio of securities are independent.
- The larger the number of securities the smaller the exposure to any particular risk.
 - “Insurance principle”
- Only issue is how many securities to hold.
- Random diversification:
 - Diversifying without looking at relevant investment characteristics.
 - Marginal risk reduction gets smaller and smaller as more securities are added.
- A large number of securities is not required for significant risk reduction.
- International diversification benefits.

Portfolio Risk and Diversification (see slide # 19)

Markowitz Diversification

- Non-random diversification:
 - Active measurement and management of portfolio risk.
 - Investigate relationships between portfolio securities before making a decision to invest.
 - Takes advantage of expected return and risk for individual securities and how security returns move together.

Measuring Portfolio Risk

- Needed to calculate risk of a portfolio:
 - Weighted individual security risks.
 - Calculated by a weighted variance using the proportion of funds in each security.
 - For security i : $(w_i \times \sigma_i)^2$
 - Weighted co-movements between returns.
 - Return co-variances are weighted using the proportion of funds in each security.
 - For securities i, j : $2w_i w_j \times \sigma_{ij}$

Correlation Coefficient

- Statistical measure of association.

- ρ_{mn} = correlation coefficient between securities m and n
 - $\rho_{mn} = +1.0$ = perfect positive correlation
 - $\rho_{mn} = -1.0$ = perfect negative (inverse) correlation
 - $\rho_{mn} = 0.0$ = zero correlation
- When does diversification pay?
 - With perfectly positive correlated securities?
 - Risk is a weighted average, therefore there is no risk reduction.
 - With zero correlation correlation securities?
 - With perfectly negative correlated securities?

Covariance

- Absolute measure of association:
 - Not limited to values between -1 and +1
 - Sign interpreted the same as correlation.
 - Correlation coefficient and covariance are related by the following equations:

Calculating Portfolio Risk

- Encompasses three factors:
 - Variance (risk) of each security.
 - Covariance between each pair of securities.
 - Portfolio weights for each security.
- Goal: select weights to determine the minimum variance combination for a given level of expected return.
- Generalizations:
 - The smaller the positive correlation between securities, the better.
 - Covariance calculations grow quickly.
 - $n(n-1)$ for n securities
 - As the number of securities increases:
 - The importance of covariance relationships increases.
 - The importance of each individual security's risk decreases.

Simplifying Markowitz Calculations

- Markowitz full-covariance model:
 - Requires a covariance between the returns of all securities in order to calculate portfolio variance.
 - $n(n-1)/2$ set of co-variances for n securities.
- Markowitz suggests using an index to which all securities are related to simplify.

An Efficient Portfolio

- Smallest portfolio risk for a given level of expected return.
- Largest expected return for a given level of portfolio risk.
- From the set of all possible portfolios:
 - Only locate and analyze the subset known as the efficient set.
 - Lowest risk for given level of return.
- All other portfolios in attainable set are dominated by efficient set.
- Global minimum variance portfolio.
 - Smallest risk of the efficient set of portfolios.
- Efficient set:
 - Part of the efficient frontier with greater risk than the global minimum variance portfolio.

Lecture # 35

Portfolio Selection

- Diversification is key to optimal risk management.
- Analysis required because of the infinite number of portfolios of risky assets.
- How should investors select the best risky portfolio?
- How could riskless assets be used?

Building a Portfolio

- **Step 1:**
 - Use the Markowitz portfolio selection model to identify optimal combinations.
 - Estimate expected returns, risk, and each covariance between returns.
- **Step 2:**

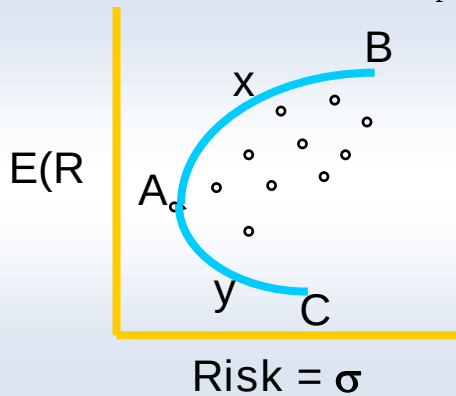
- Choose the final portfolio based on your preferences for return relative to risk.

Portfolio Theory

- Optimal diversification takes into account all available information.
- Assumptions in portfolio theory:
 - A single investment period (one year).
 - Liquid position (no transaction costs).
 - Preferences based only on a portfolio's expected return and risk.

Efficient Portfolios

- Efficient frontier or Efficient set (curved line from A to B).
- Global minimum variance portfolio (represented by point A).



Selecting an Optimal Portfolio of Risky Assets

- Assume investors are risk averse.
- Indifference curves help select from efficient set.
 - Description of preferences for risk and return.
 - Portfolio combinations which are equally desirable.
 - Greater slope implies greater the risk aversion.
- Markowitz portfolio selection model:
 - Generates a frontier of efficient portfolios which are equally good.
 - Does not address the issue of riskless borrowing or lending.
 - Different investors will estimate the efficient frontier differently.
 - Element of uncertainty in application.

The Single Index Model (slide 20)

- Relates returns on each security to the returns on a common index, such as the S&P 500 Stock Index.
- Expressed by the following equation:
- **Divides return into two components:**
 - a unique part, a_i
 - a market-related part, $b_i R_M$
 - b measures the sensitivity of a stock to stock market movements.
 - If securities are only related in their common response to the market.
 - Securities covary together only because of their common relationship to the market index.
 - Security covariance depend only on market risk and can be written as:
- Single index model helps split a security's total risk into:
 - Total risk = market risk + unique risk (slide 22)

- Multi-Index models as an alternative:
 - Between the full variance-covariance method of Markowitz and the single-index model.

Selecting Optimal Asset Classes

- Another way to use Markowitz model is with asset classes.
 - Allocation of portfolio assets to broad asset categories.
 - Asset class rather than individual security decisions most important for investors.
 - Different asset classes offers various returns and levels of risk.
 - Correlation coefficients may be quite low.

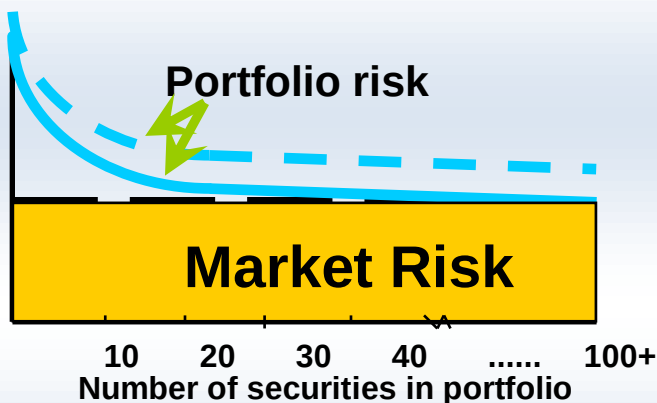
Asset Allocation

- Decision about the proportion of portfolio assets allocated to equity, fixed-income, and money market securities.
 - Widely used application of Modern Portfolio Theory.
 - Because securities within asset classes tend to move together, asset allocation is an important investment decision.
 - Should consider international securities, real estate, and U.S. Treasury TIPS.

Balanced Portfolio

Implications of Portfolio Selection

- Investors should focus on risk that cannot be managed by diversification.
- Total risk = systematic (non-diversifiable) risk + nonsystematic (diversifiable) risk
 - Systematic risk:
 - Variability in a security's total returns directly associated with economy-wide events.
- Common to virtually all securities.
 - Both risk components can vary over time.
 - Affects number of securities needed to diversify.



Lecture # 36

Asset Pricing Models

Capital Asset Pricing Model

- CAPM focuses on the equilibrium relationship between the risk and expected return on risky assets.
- It builds on Markowitz portfolio theory.
- Each investor is assumed to diversify his or her portfolio according to the Markowitz model.

CAPM Assumptions

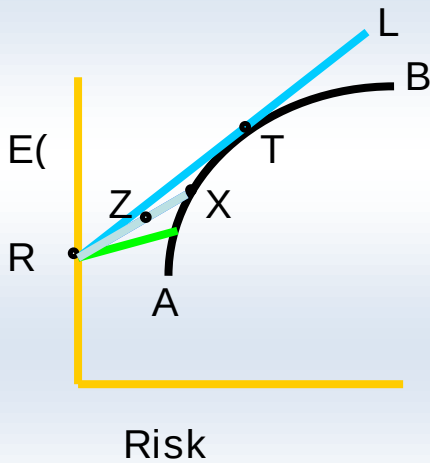
- CAPM focuses on the equilibrium relationship between the risk and expected return on risky assets.
- It builds on Markowitz portfolio theory.
- Each investor is assumed to diversify his or her portfolio according to the Markowitz model.
- No transaction costs, no personal income taxes, no inflation.
- No single investor can affect the price of a stock.
- Capital markets are in equilibrium.

Borrowing and Lending Possibilities

- Risk free assets:
 - Certain-to-be-earned expected return and a variance of return of zero.
 - No correlation with risky assets.
 - Usually proxied by a Treasury security.
 - Amount to be received at maturity is free of default risk, known with certainty.
- Adding a risk-free asset extends and changes the efficient frontier.

Risk-Free Lending

- Riskless assets can be combined with any portfolio in the efficient set AB.
 - Z implies lending
- Set of portfolios on line RF to T dominates all portfolios below it.



Impact of Risk-Free Lending

- If w_{RF} placed in a risk-free asset.
 - Expected portfolio return:
 - Risk of the portfolio:
- Expected return and risk of the portfolio with lending is a weighted average.

Borrowing Possibilities

- Investor no longer restricted to own wealth.
- Interest paid on borrowed money.
 - Higher returns sought to cover expense.
 - Assume borrowing at RF.
- Risk will increase as the amount of borrowing increases.
 - Financial leverage

The New Efficient Set

- Risk-free investing and borrowing creates a new set of expected return-risk possibilities.
- Addition of risk-free asset results in:
 - A change in the efficient set from an arc to a straight line tangent to the feasible set without the riskless asset.
 - Chosen portfolio depends on investor's risk-return preferences.

Portfolio Choice

- The more conservative the investor the more is placed in risk-free lending and the less borrowing.
- The more aggressive the investor the less is placed in risk-free lending and the more borrowing.
 - Most aggressive investors would use leverage to invest more in portfolio T.

Market Portfolio

- Most important implication of the CAPM:
 - All investors hold the same optimal portfolio of risky assets.
 - The optimal portfolio is at the highest point of tangency between RF and the efficient frontier.
 - The portfolio of all risky assets is the optimal risky portfolio.
 - Called the market portfolio.

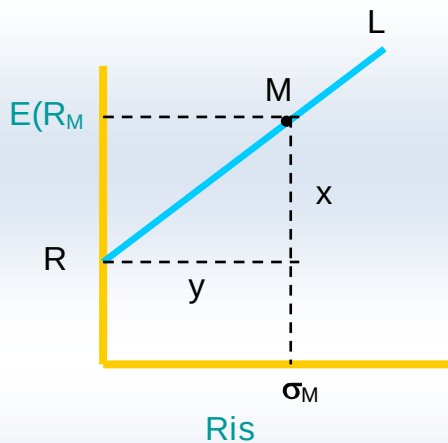
Characteristics of the Market Portfolio

- All risky assets must be in portfolio, so it is completely diversified.
 - Includes only systematic risk.
- All securities included in proportion to their market value.
- Unobservable but proxied by S&P 500.
- Unobservable but proxied by KSE 100 index.
- Contains worldwide assets.
 - Financial and real assets.

Capital Market Line

- Line from RF to L is capital market line (CML).

- x = risk premium $= E(R_M) - R_F$
- y = risk $= \sigma_M$
- Slope $= x/y$
 $= [E(R_M) - R_F] / \sigma_M$
- y-intercept $= R_F$



The Separation Theorem

- Investors use their preferences (reflected in an indifference curve) to determine their optimal portfolio.
- Separation Theorem:
 - The investment decision, which risky portfolio to hold, is separate from the financing decision.
 - Allocation between risk-free asset and risky portfolio separate from choice of risky portfolio, T .
- All investors:
 - Invest in the same portfolio.
 - Attain any point on the straight line R_F - T - L by either borrowing or lending at the rate R_F , depending on their preferences.
- Risky portfolios are not tailored to each individual's taste.

Capital Market Line

- Slope of the CML is the market price of risk for efficient portfolios, or the equilibrium price of risk in the market.
- Relationship between risk and expected return for portfolio P (Equation for CML): slide 29 formula

Security Market Line

- CML Equation only applies to markets in equilibrium and efficient portfolios.
- The Security Market Line depicts the tradeoff between risk and expected return for individual securities.
- Under CAPM, all investors hold the market portfolio.
 - How does an individual security contribute to the risk of the market portfolio?
- A security's contribution to the risk of the market portfolio is based on beta.
- Equation for expected return for an individual stock: slide 31
- Beta = 1.0 implies as risky as market.
- Securities A and B are more risky than the market.
 - Beta > 1.0
- Security C is less risky than the market.
 - Beta < 1.0 (slide 32)
- Beta measures systematic risk:
 - Measures relative risk compared to the market portfolio of all stocks.
 - Volatility different than market.
- All securities should lie on the SML:
 - The expected return on the security should be only that return needed to compensate for systematic risk.

Lecture # 37

CAPM's Expected Return-Beta Relationship

- Required rate of return on an asset (k_i) is composed of;
 - Risk-free rate (R_F)
 - Risk premium ($\beta_i [E(R_M) - R_F]$)
 - Market risk premium adjusted for specific security.

$$k_i = R_F + \beta_i [E(R_M) - R_F]$$

- The greater the systematic risk, the greater the required return.

Estimating the SML

- Treasury Bill rate used to estimate R_F .
- Expected market return unobservable.
 - Estimated using past market returns and taking an expected value.
- Estimating individual security betas difficult.
 - Only company-specific factor in CAPM.
 - Requires asset-specific forecast.

Capital Market Line

Estimating Beta

- Market model
- Characteristic line

Estimating Beta

- Market model:
 - Relates the return on each stock to the return on the market, assuming a linear relationship.

$$R_i = \alpha_i + \beta_i R_M + e_i$$

- Characteristic line:
 - Line fit to total returns for a security relative to total returns for the market index.

Betas

How Accurate Are Beta Estimates?

- Betas change with a company's situation.
 - Not stationary over time.
- Estimating a future beta.
 - May differ from the historical beta.
- R_M represents the total of all marketable assets in the economy.
 - Approximated with a stock market index.
 - Approximates return on all common stocks.
- No one correct number of observations and time periods for calculating beta.
- The regression calculations of the true α and β from the characteristic line are subject to error in estimation.
- Portfolio betas more reliable than individual security betas.

Assessments

Arbitrage Pricing Theory

- Based on the Law of One Price:
 - Two otherwise identical assets cannot sell at different prices.
 - Equilibrium prices adjust to eliminate all arbitrage opportunities.
- Unlike CAPM, APT does not assume.
 - Single-period investment horizon, absence of personal taxes, riskless borrowing or lending, mean-variance decisions.

Factors

- APT assumes returns generated by a factor model.
- Factor Characteristics:
 - Each risk must have a pervasive influence on stock returns.
 - Risk factors must influence expected return and have non-zero prices.
 - Risk factors must be unpredictable to the market.

APT Model

- Most important are the deviations of the factors from their expected values.
- The expected return-risk relationship for the APT can be described as:

$$E(R_i) = R_F + b_{i1} (\text{risk premium for factor 1}) + b_{i2} (\text{risk premium for factor 2}) + \dots + b_{in} (\text{risk premium for factor } n)$$

Problems with APT

- Factors are not well specified.
 - To implement the APT model, need the factors that account for the differences among security returns.
 - CAPM identifies market portfolio as single factor.

- Neither CAPM or APT has been proven superior.
 - Both rely on unobservable expectations

Arbitrage Pricing Theory

Portfolio Management

- Involves decisions that must be made by every investor whether an active or passive investment approach is followed.
- Relationships between various investment alternatives must be considered if an investor is to hold an optimal portfolio.

Portfolio Management as a Process

- Definite structure everyone can follow.
- Integrates a set of activities in a logical and orderly manner.
- Continuous and systematic.
- Encompasses all portfolio investments.
- With a structured process, anyone can execute decisions for investor.
- Objectives, constraints, and preferences are identified.
 - Leads to explicit investment policies.
- Strategies developed and implemented.
- Market conditions, asset mix, and investor circumstances are monitored.
- Portfolio adjustments are made as necessary.

Individual vs. Institutional Investors

Institutional Investors	Individual Investors
Maintain relatively constant profile over time.	Life stage matters
Legal and regulatory constraints.	Risk defined as “losing money”.
Well-defined and effective policy is critical.	Characterized by personalities.

Individual Investors

- Goals are important.
- Tax management is important part of decisions.

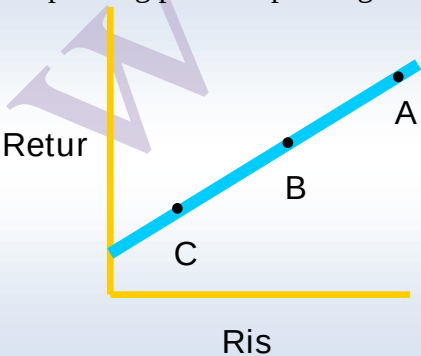
Institutional Investors

- Primary reason for establishing a long-term investment policy for institutional investors:
 - Prevents arbitrary revisions of a soundly designed investment policy.
 - Helps portfolio manager to plan and execute on a long-term basis.
 - Short-term pressures resisted.

Lecture # 38

Life Cycle Approach

- Risk/return position at various life cycle stages:
A: Accumulation phase - early career.
B: Consolidation phase - mid-to late career.
C: Spending phase - spending and gifting.



Formulate Investment Policy

- Investment policy summarizes the objectives, constraints, and preferences for the investor.
- Information needed:
 - Objectives:
 - Return requirements and risk tolerance.

- Constraints and Preferences:
 - Liquidity, time horizon, laws and regulations, taxes, unique preferences, circumstances.
- Investment policy should contain a statement about inflation adjusted returns.
 - Clearly a problem for investors.
 - Common stocks are not always an inflation hedge.
- Unique needs and circumstances.
 - May restrict certain asset classes.
- Risky Assets
- Riskless Assets
- Preferences
- No. of assets / portfolios based on personal constraints or preferences.
- Constraints and Preferences:
 - Time horizon:
 - Objectives may require specific planning horizon.
 - Liquidity needs:
 - Investors should know future cash needs.
 - Tax considerations:
 - Ordinary income vs. capital gains.
 - Retirement programs offer tax sheltering.

Legal and Regulatory Requirements

- Prudent Man Rule:
 - Followed in fiduciary responsibility.
 - Interpretation can change with time and circumstances.
 - Standard applied to individual investments rather than the portfolio as a whole.
- Diversification and standards applied to entire portfolio.

Portfolio

Capital Market Expectations

- Macro factors:
 - Expectations about the capital markets.
- Micro factors:
 - Estimates that influence the selection of a particular asset for a particular portfolio.
- Rate of return assumptions:
 - Make them realistic.
 - Study historical returns carefully.
 - Qualitative & quantitative data based on historical data.

Rate of Return Assumptions

- How much influence should recent stock market returns have?
 - Mean reversion arguments.
 - Stock returns involve considerable risk.
 - Probability of 10% return is 50% regardless of the holding period.
 - Probability of >10% return decreases over longer investment horizons.
 - Expected returns are not guaranteed.

Constructing the Portfolio

- Use investment policy and capital market expectations to choose portfolio of assets.
 - Define securities eligible for inclusion in a particular portfolio.
 - Use an optimization procedure to select securities and determine the proper portfolio weights.
 - Markowitz provides a formal model.
- Numbers game
- Capital gains
- Returns
- Required rate of return
- Time horizon

Asset Allocation

- Involves deciding on weights for cash, bonds, and stocks.
 - Most important decision:
 - Differences in allocation cause differences in portfolio performance.
- Factors to consider:

- Return requirements, risk tolerance, time horizon, age of investor.
- Strategic asset allocation:
 - Simulation procedures used to determine likely range of outcomes associated with each asset mix.
 - Establishes long-run strategic asset mix.
- Tactical asset allocation:
 - Changes is asset mix driven by changes in expected returns.
 - Market timing approach.

Monitoring Conditions and Circumstances

- Investor circumstances can change for several reasons;
 - Wealth changes affect risk tolerance.
 - Investment horizon changes.
 - Liquidity requirement changes.
 - Tax circumstance changes.
 - Regulatory considerations.
 - Unique needs and circumstances.

Portfolio Adjustments

- Portfolio not intended to stay fixed.
- Key is to know when to rebalance.
- Rebalancing cost involves:
 - Brokerage commissions.
 - Possible impact of trade on market price.
 - Time involved in deciding to trade.
- Cost of not rebalancing involves holding unfavorable positions.

Performance Measurement

- Allows measurement of the success of portfolio management.
- Key part of monitoring strategy and evaluating risks.
- Important for:
 - Those who employ a manager.
 - Those who invest personal funds.
- Find reasons for success or failure.

Evaluation of the Investment Process

How Should Portfolio Performance Be Evaluated?

- Bottom line” issue in investing.
- Is the return after all expenses adequate compensation for the risk?
- What changes should be made if the compensation is too small?
- Performance must be evaluated before answering these questions.

Lecture # 39

Considerations

- Without knowledge of risks taken, little can be said about performance.
 - Intelligent decisions require an evaluation of risk and return.
 - Risk-adjusted performance best.
- Relative performance comparisons.
 - Benchmark portfolio must be legitimate alternative that reflects objectives.
- Evaluation of portfolio manager or the portfolio itself?
 - Portfolio objectives and investment policies matter.
 - Constraints on managerial behavior affect performance.
- How well-diversified during the evaluation period?
 - Adequate return for diversifiable risk?
- Age of investors
- Financial liquidity strength of the investors
- The cash flow requirement of the investors
- The risk tolerance of the investors

Return Measures

- Change in investor’s total wealth over an evaluation period;

$$(VE - VB) / VB$$

VE =ending portfolio value

VB =beginning portfolio value

- Assumes no funds added or withdrawn during evaluation period.
 - If not, timing of flows important.

Return Measures

- Portfolios is the mean of enhancing the somebody's capital gains, capital wealth and his financial status as well.
- Dollar-weighted returns
- Time-weighted returns
- Dollar-weighted returns:
 - Captures cash flows during the evaluation period.
 - Equivalent to internal rate of return.
 - Equates initial value of portfolio (investment) with cash inflows or outflows and ending value of portfolio.

$(VE - VB) / VB$

- Cash flow effects make comparisons to benchmarks inappropriate.
- Time-weighted returns:
 - Captures cash flows during the evaluation period and permits comparisons with benchmarks.
 - Calculate a return relative for each time period defined by a cash inflow or outflow.
 - Use each return relative to calculate a compound rate of return for the entire period.

Which Return Measure Should Be Used?

- Dollar- and Time-weighted Returns can give different results;
 - Dollar-weighted returns appropriate for portfolio owners.
 - Time-weighted returns appropriate for portfolio managers.
 - No control over inflows, outflows.
 - Independent of actions of client.
- One could look at time-weighted returns / results.

Risk Measures

- Risk differences cause portfolios to respond differently to market changes.
- Total risk measured by the standard deviation of portfolio returns.
- Non-diversifiable risk measured by a security's beta.
 - Estimates may vary, be unstable, and change over time.

Risk-Adjusted Performance

- The Sharpe reward-to-variability ratio:
 - Benchmark based on the ex post capital market line.

=Average excess return / total risk

- Risk premium per unit of risk.
- The higher, the better the performance.
- Provides a ranking measure for portfolios.
- The Treynor reward-to-volatility ratio:
 - Distinguishes between total and systematic risk.

- =Average excess return / market risk
- Risk premium per unit of market risk.
- The higher, the better the performance.
- Implies a diversified portfolio.

RVAR or RVOL?

- Depends on the definition of risk:
 - If total risk best, use RVAR.
 - If systematic risk best, use RVOL.
 - If portfolios perfectly diversified, rankings based on either RVAR or RVOL are the same.
 - Differences in diversification cause ranking differences.
 - RVAR captures portfolio diversification

Measuring Diversification

- How correlated are portfolio's returns to market portfolio?
 - R2 from estimation of;

$R_{pt} - R_{Ft} = a_p + b_p [R_{Mt} - R_{Ft}] + e_{pt}$

- R^2 is the coefficient of determination.
- Excess return form of characteristic line.
- The lower the R^2 , the greater the diversifiable risk and the less diversified.

Jensen's Alpha

- The estimated α coefficient in

$$R_{pt} - R_{ft} = \alpha + \beta_p [R_{Mt} - R_{ft}] + \epsilon_{pt}$$

is a means to identify superior or inferior portfolio performance.

- CAPM implies α is zero.
- Measures contribution of portfolio manager beyond return attributable to risk.
- If $\alpha > 0$ (< 0 , $= 0$), performance superior (inferior, equals) to market, risk-adjusted.

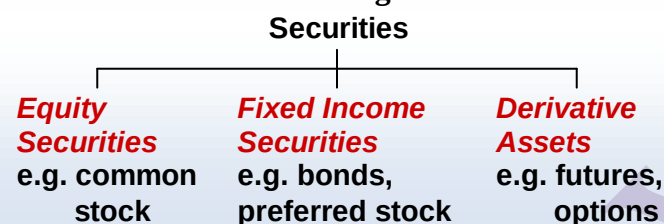
Measurement Problems

- Performance measures based on CAPM and its assumptions.
 - Riskless borrowing?
 - What should market proxy be?
 - If not efficient, benchmark error.
 - Global investing increases problem.
- How long an evaluation period?
 - One could look at a 10 year period.

Other Evaluation Issues

- Performance attribution seeks an explanation for success or failure.
 - Analysis of investment policy and asset allocation decision.
 - Analysis of industry and security selection.
 - Benchmark (bogey) selected to measure passive investment results.
 - Differences due to asset allocation, market timing, security selection.

There are three broad categories of securities.



Three Reasons for Investing

People invest to ...

- **Earn capital gains:**
 - Appreciation refers to an increase in the value of an investment.
- **Supplement their income:**
 - Dividends and Interest.
- **Experience the excitement** of the investment process.

Lecture # 40

What are Derivatives?

- **Derivatives** are instruments whose values depend on the value of other more basic underlying variables. Their value *derives* from some other asset.
 - A derivative is a financial instrument whose return is derived from the return on another instrument.

The underlying assets could be:

- Financial Assets
- Commodities
- Real Assets

The Rationale for Derivative Assets

- The first organized derivatives exchange (in the United States) was developed in order to bring stability to agricultural prices, by enabling farmers to eliminate or reduce their **price risk**.

Introduction

- There is no universally satisfactory answer to the question of “What a derivative is”?
- Often when a market participant suffers a large newsworthy loss, the term “derivatives” is used almost as if it were an explanation.
 - “anything that results in a large loss”

- “dreaded D word”
- “beef derivative”
- Futures and options markets are very useful, perhaps even essential, parts of the financial system.
- Futures and options markets have a long history of being misunderstood.

Derivatives Markets

- **Exchange traded:**
 - Contracts are standard there is virtually no credit risk.
 - Traditionally exchanges have used the open-outcry system, but increasingly they are switching to electronic trading.
- **Over-the-counter (OTC):**
 - A computer- and telephone-linked network of dealers at financial institutions, corporations, and fund managers.
 - Contracts can be non-standard and there is some small amount of credit risk.

Role of Derivative Assets

- Derivative assets:
 - The best-known derivative assets are futures and options contracts.
 - Derivatives are not all the same. Some are inherently speculative, while some are highly conservative.

Derivatives Instruments

- Futures Contracts
- Forwards Contracts
- Options
- Swaps

Futures Contract:

- Traded on exchanges.
- It is an agreement to sell or buy in future like a forward contract.

Forward Contract:

- Traded in Over-the-Counter Market.
- It is an agreement to buy or sell an asset at a certain future time for a certain price.
 - **Option contract** gives the owner of the contract a right but not an obligation to buy/sell in the future.
 - Traded both at exchanges and over-the-counter markets.
 - **Swap:**
 - It is an agreement to exchange cash flows at specified future times according to certain specified rules.

Types of Derivatives

- Options
- Futures contracts
- Swaps
- Product Characteristics

Options

- An **option** is the right to either buy or sell something at a set price, within a set period of time.
 - The right to buy is a **call option**.
 - The right to sell is a **put option**.
- You can exercise an option if you wish, but you do not have to do so.

Futures Contracts

- **Futures contracts** involve a promise to exchange a product for cash by a set delivery date.
- **Futures contracts** deal with transactions that will be made in the future.
- Are different from options in that:
 - The buyer of an option can abandon the option if he or she wishes.
 - The buyer of a **futures contract** cannot abandon the contract.

Futures Contracts Example

- The futures market deals with transactions that will be made in the future. A person who buys a December U.S. Treasury bond futures contract promises to pay a certain price for treasury bonds in December. If you buy the T-bonds today, you purchase them in the cash, or **spot, market**.

Futures Contracts

- A futures contract involves a process known as *marking to market*.
 - Money actually moves between accounts each day as prices move up and down.

- A *forward contract* is functionally similar to a futures contract, however:
 - There is no marking to market.
 - Forward contracts are not marketable.

SWAPS

- **Swaps** are arrangements in which one party trades something with another party.
- The swap market is very large, with trillions of dollars outstanding

Interest Rate Swap

In an **interest rate swap**, one firm pays a fixed interest rate on a sum of money and receives from some other firm a floating interest rate on the same sum.

Foreign Currency Swap

- In a **foreign currency swap**, two firms initially trade one currency for another.
- Subsequently, the two firms exchange interest payments, one based on a foreign interest rate and the other based on a U.S. interest rate.
- Finally, the two firms re-exchange the two currencies.

Product Characteristics

- Both options and futures contracts exist on a wide variety of assets.
 - Options trade on individual stocks, on market indexes, on metals, interest rates, or on futures contracts.
 - Futures contracts trade on products such as wheat, live cattle, gold, heating oil, foreign currency, U.S. Treasury bonds, and stock market indexes.
- The **underlying asset** is that which you have the right to buy or sell (with options) or to buy or deliver (with futures).
- **Listed derivatives** trade on an organized exchange such as the Chicago Board Options Exchange (CBOE) or the Chicago Board of Trade (CBT).
- **OTC derivatives** are customized products that trade off the exchange and are individually negotiated between two parties.
- Options are securities and are regulated by the Securities and Exchange Commission (SEC).
- Futures contracts are regulated by the Commodity Futures Trading Commission (CFTC).

Lecture # 41

Key Points About Futures

- They are settled daily.
- Closing out a futures position involves entering into an offsetting trade.
- Most contracts are closed out before maturity.

Features of Future Contracts

- Available on a wide range of underlying assets.
- Exchange traded
- Specifications need to be defined:
 - The Underlying. This can be anything from a barrel of sweet crude oil to a short term interest rate.
 - The Amount And Units Of The Underlying Asset Per Contract.
 - The Delivery Month.
 - The Last Trading Date.
 - The Grade Of The Deliverable.
 - The Type Of Settlement, either cash settlement or physical settlement.
 - The Currency in which the futures contract is quoted.
- Settled daily for profits / losses.

Delivery

- If a futures contract is not closed out before maturity, it is usually settled by delivering the assets underlying the contract.
- When there are alternatives about what is delivered, where it is delivered, and when it is delivered, the party with the short position chooses.

Margins

- A margin is cash or marketable securities deposited by an investor with his or her broker.
- The balance in the margin account is adjusted to reflect daily settlement.
- Margins minimize the possibility of a loss through a default on a contract.

Understanding Futures Markets

- Spot or cash market:
 - Price refers to item available for immediate delivery.
- Forward market:
 - Price refers to item available for delayed delivery.
- Futures market:
 - Sets features (contract size, delivery date, and conditions) for delivery.
- Futures market characteristics:
 - Centralized marketplace allows investors to trade each other.
 - Performance is guaranteed by a clearinghouse.
- Valuable economic functions:
 - Hedgers shift price risk to speculators.
 - Price discovery conveys information.
- Commodities - agricultural, metals, and energy related.
- Financials - foreign currencies as well as debt and equity instruments.
- Foreign futures markets:
 - Increased number shows the move toward globalization.
 - Markets quite competitive with US environment.

Futures Contract

- A obligation to buy or sell a fixed amount of an asset on a specified future date at a price set today.
 - Trading means that a commitment has been made between buyer and seller.
 - Position offset by making an opposite contract in the same commodity.
- Commodity Futures Trading Commission regulates trading.

Futures Exchanges

- Where futures contracts are traded.
- Voluntary, nonprofit associations, of membership.
- Organized marketplace where established rules govern conduct.
 - Funded by dues and fees for services rendered.
- Members trade for self or for others.

The Clearinghouse

- A corporation separate from, but associated with, each exchange.
- Exchange members must be members or pay a member for these services.
 - Buyers and sellers settle with clearinghouse, not with each other.
- Helps facilitate an orderly market.
- Keeps track of obligations.

The Mechanics of Trading

- Through open-outcry, seller and buyer agree to take or make delivery on a future date at a price agreed on today.
 - Short position (seller) commits a trader to deliver an item at contract maturity.
 - Long position (buyer) commits a trader to purchase an item at contract maturity.
 - Like options, futures trading a zero sum game.
- How contracts are settled?
- Contracts can be settled in two ways:
 - Delivery (less than 2% of transactions).
 - Offset: liquidation of a prior position by an offsetting transaction.
- Each exchange establishes price fluctuation limits on contracts.
- No restrictions on short selling.

Margin

- A margin is cash or marketable securities deposited by an investor with his or her broker.
- The balance in the margin account is adjusted to reflect daily settlement.
- Margins minimize the possibility of a loss through a default on a contract.

Futures (Margin)

Earnest money deposit made by both buyer and seller to ensure performance of obligations.

- Not an amount borrowed from broker.
- Each clearinghouse sets requirements.
 - Brokerage houses can require higher margin.
- Initial margin usually less than 10% of contract value.

Futures (Margin calls)

- Margin calls occur when price goes against investor.
 - Must deposit more cash or close account.
 - Position marked-to-market daily.
 - Profit can be withdrawn.

Lecture # 42

Participants in the Derivatives World

- Hedging
- Speculation
- Arbitrage

Ways Derivatives are Used

- To hedge risks
- To speculate (take a view on the future direction of the market).
- To lock in an arbitrage profit.
- To change the nature of a liability.
- To change the nature of an investment without incurring the costs of selling one portfolio and buying another.

First Type of Trader

- **Hedger:** A person who uses derivatives to reduce risk that they may face in future.

Using Future Contracts Hedging

- If someone bears an economic risk and uses the futures market to reduce that risk, the person is a *hedger*.
- Hedging is a prudent business practice and a prudent manager has a legal duty to understand and use the futures market hedging mechanism.

Using Futures Contracts

- Hedgers:
 - At risk with a spot market asset and exposed to unexpected price changes.
 - Buy or sell futures to offset the risk.
 - Used as a form of insurance.
 - Willing to forgo some profit in order to reduce risk.
 - Hedged return has smaller chance of low return but also smaller chance of high.

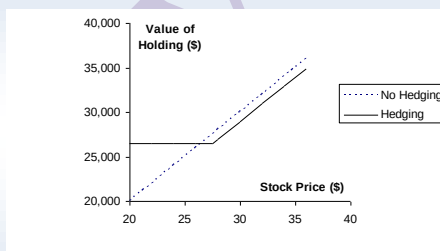
Hedging

- Short (sell) hedge:
 - Cash market inventory exposed to a fall in value.
 - Sell futures now to profit if the value of the inventory falls.
- Long (buy) hedge:
 - Anticipated purchase exposed to a rise in cost.
 - Buy futures now to profit if costs increase

Hedging Examples

- A US company will pay £10 million for imports from Britain in 3 months and decides to hedge using a long position in a forward contract.
- An investor owns 1,000 Microsoft shares currently worth \$28 per share. A two-month put with a strike price of \$27.50 costs \$1. The investor decides to hedge by buying 10 contracts.

Value of Microsoft Shares with and without Hedging



Second Type of Trader

- **Speculator:** A person who uses derivatives to bet on the future direction of the market.

Speculation

- A person or firm who accepts the risk the hedger does not want to take is a *speculator*.
- Speculators believe the potential return outweighs the risk.
- The primary purpose of derivatives markets is not speculation. Rather, they permit the transfer of risk between market participants as they desire.

Speculation Example

- An investor with \$4,000 to invest feels that amazon.com's stock price will increase over the next 2 months. The current stock price is \$40 and the price of a 2-month call option with a strike of 45 is \$2.

Third Type of Trader

- **Arbitrageur:** A person who takes two opposite positions on the same instrument in two different markets to earn a profit.
- Derivatives are excessively used for all these purposes.

Arbitrage

- *Arbitrage* is the existence of a riskless profit.
- Arbitrage opportunities are quickly exploited and eliminated.
- Persons actively engaged in seeking out minor pricing discrepancies are called *arbitrageurs*.
- Arbitrageurs keep prices in the marketplace efficient.
 - An efficient market is one in which securities are priced in accordance with their perceived level of risk and their potential return.

Arbitrage Example

- A stock price is quoted as £100 in London and \$172 in New York.
- The current exchange rate is 1.7500

Hedging Risks

- **Basis:** difference between cash price and futures price of hedged item.
 - Must be zero at contract maturity.
- **Basis risk:** the risk of an unexpected change in basis.
 - Hedging reduces risk if basis risk less than variability in price of hedged asset.
- Risk cannot be entirely eliminated.

Hedging with Stock Index Futures

- Selling futures contracts against diversified stock portfolio allows the transfer of systematic risk.
 - Diversification eliminates nonsystematic risk.
 - Hedging against overall market decline.
 - Offset value of stock portfolio because futures prices are highly correlated with changes in value of stock portfolios.

Using Futures Contracts

- **Speculators:**
 - Buy or sell futures contracts in an attempt to earn a return.
 - No prior spot market position.
 - Absorb excess demand or supply generated by hedgers.
 - Assuming the risk of price fluctuations that hedgers wish to avoid.
 - Speculation encouraged by leverage, ease of transacting, low costs.

Speculating with Stock Index Futures

- Futures effective for speculating on movements in stock market because:
 - Low transaction costs involved in establishing futures position.
 - Stock index futures prices mirror the market.
- Traders expecting the market to rise (fall) buy (sell) index futures.
- Futures contract spreads:
 - Both long and short positions at the same time in different contracts.
 - Intramarket (or calendar or time) spread.
 - Same contract, different maturities.
 - Intermarket (or quality) spread.
 - Same maturities, different contracts.
- Interested in relative price as opposed to absolute price changes.

Financial Futures

- Contracts on equity indexes, fixed income securities, and currencies.
- Opportunity to fine-tune risk-return characteristics of portfolio.
- At maturity, stock index futures settle in cash.
 - Difficult to manage delivery of all stocks in a particular index.
- At maturity, Treasury bond and Treasury bill interest rate futures settle by delivery of debt instruments.
 - If expect increase in rates, sell interest rate futures.
 - If expect decrease in rates, buy interest rate futures.
 - Increase (decrease) in interest rates will decrease (increase) spot and futures prices.

- Difficult to short bonds in spot market.

Program Trading

- Index arbitrage: a version of program trading.
 - Exploitation of price difference between stock index futures and index of stocks underlying futures contract.
 - Arbitrageurs build hedged portfolio that earns low risk profits equaling the difference between the value of cash and futures positions.

Lecture # 43

Example of a Futures Trade

An investor takes a long position in two December gold futures contracts on June 5.

- Contract size is 100 oz.
- Futures price is US\$400
- Margin requirement is US\$2,000/contract (US\$4,000 in total)
- Maintenance margin is US\$1,500/contract (US\$3,000 in total)

A Possible Outcome

Day	Futures		Daily		Cumulative	
	Price	(Loss)	Gain	(Loss)	Account	Margin
	(US\$)	(US\$)	(US\$)	(US\$)	Balance	Call
	400.00				4,000	
5-Jun	397.00	(600)		(600)	3,400	0
...	...	...	...	...	...	...
13-Jun	393.30	(740)		(1340)	2,660	+ 1,340 = 4,000
...	...	...	...	...	...	< 3,000
19-Jun	387.00	(1,260)		(2,600)	2,740	+ 1,260 = 4,000
...	...	...	...	...	...	...
26-Jun	392.30	1,060		(1,540)	5,060	0

Long & Short Hedges

- A **long futures hedge** is appropriate when you know you will purchase an asset in the future and want to lock in the price.
- A **short futures hedge** is appropriate when you know you will sell an asset in the future & want to lock in the price.

Basis Risk

- **Basis** is the difference between spot & futures.
- **Business risk** arises because of the uncertainty about the basis when the hedge is closed out.

Long Hedge

- Suppose that

F1 : Initial Futures Price

F2 : Final Futures Price

S2 : Final Asset Price

- You hedge the future purchase of an asset by entering into a long futures contract.
- Cost of Asset=S2 – (F2 – F1) = F1 + Basis

Short Hedge

- Suppose that

F1 : Initial Futures Price

F2 : Final Futures Price

S2 : Final Asset Price

- You hedge the future sale of an asset by entering into a short futures contract.
- Price Realized=S2+ (F1 – F2) = F1 + Basis

Choice of Contract

- Choose a delivery month that is as close as possible to, but later than, the end of the life of the hedge.
- When there is no futures contract on the asset being hedged, choose the contract whose futures price is most highly correlated with the asset price. This is known as cross hedging.

Reasons for Hedging an Equity Portfolio

- Desire to be out of the market for a short period of time.
- Hedging may be cheaper than selling the portfolio and buying it back.
- Desire to hedge systematic risk.
- Appropriate when you feel that you have picked stocks that will outperform the market.

Hedging Price of an Individual Stock

- Similar to hedging a portfolio.
- Does not work as well because only the systematic risk is hedged.
- The unsystematic risk that is unique to the stock is not hedged.

Hedge Equity Returns

- May want to be out of the market for a while.
- Hedging avoids the costs of selling and repurchasing the portfolio.
- Suppose stocks in your portfolio have an above average EPS, but you feel they have been chosen well and will outperform the market in both good and bad times. Hedging ensures that the return you earn is the risk-free return plus the excess return of your portfolio over the market.

Rolling The Hedge Forward

- We can use a **series of futures** contracts to increase the life of a hedge.
- Each time we switch **from one** futures contract **to another** we incur a type of basis risk.

Simple Valuation of Futures and Forward Contracts

S_0 : Spot price today

F_0 : Futures or forward price today

T : Time until delivery date

r : Risk-free interest rate for maturity T

Gold: An Arbitrage Opportunity?

- Suppose that:
 - The spot price of gold is US\$390.
 - The quoted 1-year forward price of gold is US\$425.
 - The 1-year US\$ interest rate is 5% per annum.
 - No income or storage costs for gold.
- Is there an arbitrage opportunity?

The Forward Price of Gold

If the spot price of gold is S and the futures price is for a contract deliverable in T years is F , then

$$F = S(1+r)^T$$

Where r is the 1-year (domestic currency) risk-free rate of interest.

In our examples, $S=390$, $T=1$, and $r=0.05$ so that

$$F = 390(1+0.05)^1 = 409.50$$

Uses of Derivatives

- Risk management
- Income generation
- Financial engineering
- **Risk management**: The equity manager's market risk or the bond manager's interest rate risk is similar to the farmer's price risk.
- **Risk transfer**: Derivatives provide a means for risk to be transferred from one person to some other market participant who, for a price, is willing to bear it.
- Derivatives may provide **financial leverage**.

Risk Management

- The hedger's primary motivation is risk management.
- Someone who is bullish believes prices are going to rise.
- Someone who is bearish believes prices are going to fall.
- We can tailor our risk exposure to any points we wish along a bullish / bearish sentiments of the market.

Falling prices **Flat market** **Rising prices**
expected **expected** **expected**



Bearish Neutral Bullish

Increasing **Bearishness**

Increasing **Bullishness**

For All Users of Derivatives

- Risk must be quantified and risk limits set.
- Exceeding risk limits not acceptable even when profit results.

www.vuzs.net

<http://groups.google.com/group/vuzs>

- Be diversified.
- Scenario analysis is important

Uses of Derivatives

- Risk management
- Income generation
- Financial engineering
- Writing Calls

Lecture # 44

Uses of Derivatives

Income generation : Some people use derivatives as a means of generating additional income from their investment portfolio.

- Writing a *covered call* is a way to generate income.
 - Involves giving someone the right to purchase your stock at a set price in exchange for an up-front fee (the option premium) that is yours to keep no matter what happens.
- Writing calls is especially popular during a flat period in the market or when prices are trending downward.

Financial Engineering

- **Financial engineering** refers to the practice of using derivatives as building blocks in the creation of some specialized product.
- Derivatives can be stable or volatile depending on how they are combined with other assets
- **Financial engineers:**
 - Select from a wide array of puts, calls futures, and other derivatives.
 - Know that derivatives are neutral products (neither inherently risky nor safe).

Effective Study of Derivatives

- All financial institutions can make some productive use of derivative assets.
 - Investment houses.
 - Asset-liability managers at banks.
 - Pension fund managers.
 - Mortgage officers.

What are Options

- **Call Options:**
 - A call option gives its owner the right to buy; it is not a promise to buy.
 - For example, a store holding an item for you for a fee is a call option.
- **Put Options:**
 - A put option gives its owner the right to sell; it is not a promise to sell.
 - For example, a lifetime money back guarantee policy on items sold by a company is an embedded put option.

Basic Option Characteristics

- The option *premium* is the amount you pay for the option.
- Exchange-traded options are *fungible*.
 - For a given company, all options of the same type with the same expiration and striking price are identical.
- The *striking price* of an option is its predetermined transaction price.

Standardized Option Characteristics

- **Expiration dates:**
 - The Saturday following the third Friday of certain designated months for most options.
- **Striking price:**
 - The predetermined transaction price, in multiples of \$2.50 or \$5, depending on current stock price.
- **Underlying Security:**
 - The security the option gives you the right to buy or sell.
 - Both puts and calls are based on 100 shares of the underlying security.

Where Options Come From

- Unlike more familiar securities, there is no set number of put or call options.
 - The number in existence changes every day.

Some Terminology

- **Open interest:**

The total number of contracts outstanding.

- Equal to number of long positions or number of short positions.

- **Settlement price:**

The price just before the final bell each day .

- Used for the daily settlement process.

- **Volume of trading:**

The number of trades in 1 day.

- In-the-money options have a positive cash flow if exercised immediately.
 - Call options: $S > E$
 - Put options: $S < E$
- Out-of-the-money options should not be exercised immediately.
 - Call options: $S < E$
 - Put options: $S > E$
- Intrinsic value is the value realized from immediate exercise.
 - Call options: maximum ($S - E$ or 0)
 - Put options: maximum ($E - S$ or 0)
- Prior to option maturity, option premiums exceed intrinsic value.
 - Time value = Option price - Intrinsic value
 - =seller compensation for risk
- Call (Put): Buyer has the right but not the obligation to purchase (sell) a fixed quantity from (to) the seller at a fixed price before a certain date.
 - Exercise (strike) price: “fixed price”
 - Expiration (maturity) date: “certain date”
- Option premium or price: paid by buyer to the seller to get the “right”.

Why Options Markets?

- Financial derivative securities: derive all or part of their value from another (underlying) security.
- Options are created by investors, sold to other investors.
- Why trade these indirect claims?
 - Expand investment opportunities, lower cost, increase leverage.

Why Options Are a Good Idea

- Increased risk
- Instantaneous information
- Portfolio risk management
- Risk transfer
- Financial leverage
- Income generation

How Options Work

- Call buyer (seller) expects the price of the underlying security to increase (decrease or stay steady).
- Put buyer (seller) expects the price of the underlying security to decrease (increase or stay steady).
- At option maturity:
 - Option may expire worthless, be exercised, or be sold.

Options Trading

- Option exchanges are continuous primary and secondary markets.
 - Chicago Board Options Exchange largest.
- Standardized exercise dates, exercise prices, and quantities.
 - Facilitates offsetting positions through Options Clearing Corporation.
 - OCC is guarantor, handles deliveries

Opening and Closing Transactions

- The first trade someone makes in a particular option is an *opening transaction* for that person.
- When the individual subsequently closes that position out with a second trade, this latter trade is a *closing transaction*.
- When someone buys an option as an opening transaction, the owner of an option will ultimately do one of three things with it:
 - Sell it to someone else
 - Let it expire
 - Exercise it
- For example, buying a ticket to an athletic event.

- When someone sells an option as an opening transaction, this is called *writing the option*.
 - No matter what the owner of an option does, the writer of the option keeps the option premium that he or she received when it was sold.

The Role of the Options Clearing Corporation (OCC)

- The *Options Clearing Corporation (OCC)* contributes substantially to the smooth operation of the options market.
 - It positions itself between every buyer and seller and acts as a guarantor of all option trades.
 - It sets minimum capital requirements and provides for the efficient transfer of funds among members as gains or losses occur

Where and How Options Trade

- Exchanges
- Over-the-counter options
- Standardized option characteristics
- Other listed options
- Trading mechanics

Exchanges

- Major options exchanges in the U.S.:
 - Chicago Board Options Exchange (CBOE)
 - American Stock Exchange (AMEX)
 - Philadelphia Stock Exchange (Philly)
 - Pacific Stock Exchange (PSE)
- Foreign options exchanges also exist.

Over-the-Counter Options

- With an *over-the-counter* option:
 - Institutions enter into “private” option arrangements with brokerage firms or other dealers.
 - The striking price, life of the option, and premium are negotiated between the parties involved.
- *Over-the-counter* options are subject to counterparty risk and are generally not fungible.

Some Exotic Options

- **As-You-Like-It Option:**
 - The owner can decide whether it is a put or a call by a certain date.
- **Barrier Option:**
 - Created or cancelled if a pre-specified price level is touched.
- **Forward Start Option:**
 - Paid for now, with the option becoming effective at a future date.

Other Listed Options

- *Long-Term Equity Anticipation Security (LEAP):*
 - Options similar to ordinary listed options, except they are longer term.
 - May have a life up to 39 months.

Lecture # 45 (Over all Revision)

1. Investing
2. Investment Alternatives
3. Category of Stocks
4. Types Of Orders
5. Types of Accounts
6. Fundamental Analysis
7. Sector / Industry Analysis
8. Balance Sheet
9. Income Statement
10. Statement of Cash Flow
11. Multiples
12. Fundamental vs. Technical
13. Charts
14. Basic Technical Tools
 - Trend Lines
 - Moving Averages
 - Price Patterns
 - Indicators

Cycles

15. Indirect Investments Investment Companies
16. Passive Stock Strategies
17. Efficient Market Hypothesis
18. Index
19. Alternative Investments Non-marketable Financial Assets
20. Money Market Securities
21. Bond Prices
22. Bond Ratings

Investment grade securities:

- a. Rated AAA, AA, A, BBB
- b. Typically, institutional investors are confined to bonds in these four categories.

Speculative securities:

- c. Rated BB, B, CCC, C
- d. Significant uncertainties.
- e. C rated bonds are not paying interest.

23. Risk
24. Risk Types
25. Risk Sources
26. Portfolio Theory
27. Markowitz Diversification
28. An Efficient Portfolio
29. Capital Asset Pricing Model
30. Portfolio Choice
31. Market Portfolio
32. How Accurate Are Beta Estimates?
33. Arbitrage Pricing Theory
34. Portfolio Management
35. Performance Measurement
36. What are Derivatives?
37. Derivatives Instruments
38. Understanding Futures Markets
39. Participants In the Derivatives World
40. Ways Derivatives are Used
41. Using Futures Contracts
42. Speculation
43. Third Type of Trader
44. Uses of Derivatives
45. Uses of Derivatives
46. Financial Engineering
47. What are Options
48. Basic Option Characteristics

**Always Remember Our Group in Your Prayers:
(Humera Fazal, Farhana Burhan, Ayesha
Masood, Shaista, Zulfiqar Gill & Naveed)**